Exercise 1: Inventory Management System

Product.java

```
public class Product {
   int productId;
   String productName;
   int quantity;
   double price;

   public Product(int productId, String productName, int quantity, double price) {
      this.productId = productId;
      this.productName = productName;
      this.quantity = quantity;
      this.price = price;
   }

   @Override
   public String toString() {
      return productId + " " + productName +
            " Qty:" + quantity + " Price:" + price;
   }
}
```

InventoryManagement.java

```
import java.util.HashMap;

public class InventoryManagement {

   static HashMap<Integer, Product> inventory = new HashMap<>();

   public static void addProduct(Product p) {
      inventory.put(p.productId, p);
   }

   public static void updateProduct(int id, int quantity) {
      if (inventory.containsKey(id)) {
         inventory.get(id).quantity = quantity;
      }
   }

   public static void deleteProduct(int id) {
```

```
        inventory.remove(id);
    }

    public static void display() {
        for (Product p : inventory.values()) {
            System.out.println(p);
        }
    }

    public static void main(String[] args) {
        addProduct(new Product(101, "Laptop", 10, 50000));
        addProduct(new Product(102, "Mouse", 50, 500));

        display();

        updateProduct(101, 15);
        deleteProduct(102);

        System.out.println("\nAfter Update/Delete");
        display();
    }
}
```

Complexity

- Add → O(1)
- Update → O(1)
- Delete → O(1)

Exercise 2: E-Commerce Search Function

Product.java

```
public class Product {
    int productId;
    String productName;
    String category;

    public Product(int id, String name, String category) {
        this.productId = id;
        this.productName = name;
        this.category = category;
```

```
    }
}
```

SearchDemo.java

```
public class SearchDemo {

    public static int linearSearch(Product[] products, String key) {
        for (int i = 0; i < products.length; i++) {
            if (products[i].productName.equalsIgnoreCase(key))
                return i;
        }
        return -1;
    }

    public static int binarySearch(Product[] products, String key) {
        int low = 0, high = products.length - 1;

        while (low <= high) {
            int mid = (low + high) / 2;
            int result = products[mid].productName.compareToIgnoreCase(key);

            if (result == 0)
                return mid;
            else if (result < 0)
                low = mid + 1;
            else
                high = mid - 1;
        }
        return -1;
    }

    public static void main(String[] args) {

        Product[] products = {
            new Product(1,"Keyboard","Electronics"),
            new Product(2,"Laptop","Electronics"),
            new Product(3,"Mouse","Electronics")
        };

        System.out.println(
            "Linear Search: " +
```

```
        linearSearch(products,"Laptop"));

      System.out.println(
        "Binary Search: " +
        binarySearch(products,"Laptop"));
  }
}
```

Complexity

- Linear Search → O(n)
- Binary Search → O(log n)

Exercise 3: Sorting Customer Orders

Order.java

```
public class Order {
  int orderId;
  String customerName;
  double totalPrice;

  public Order(int id, String name, double price) {
    this.orderId = id;
    this.customerName = name;
    this.totalPrice = price;
  }

  public String toString() {
    return orderId + " " + customerName +
        " " + totalPrice;
  }
}
```

SortOrders.java

```
public class SortOrders {

  static void bubbleSort(Order[] arr) {
    int n = arr.length;

    for(int i=0;i<n-1;i++) {
      for(int j=0;j<n-i-1;j++) {
        if(arr[j].totalPrice > arr[j+1].totalPrice) {
```

```
            Order temp = arr[j];
            arr[j]=arr[j+1];
            arr[j+1]=temp;
          }
       }
    }
}

static int partition(Order[] arr,int low,int high) {
    double pivot = arr[high].totalPrice;
    int i = low-1;

    for(int j=low;j<high;j++) {
       if(arr[j].totalPrice < pivot) {
          i++;
          Order temp=arr[i];
          arr[i]=arr[j];
          arr[j]=temp;
       }
    }

    Order temp=arr[i+1];
    arr[i+1]=arr[high];
    arr[high]=temp;

    return i+1;
}

static void quickSort(Order[] arr,int low,int high) {
    if(low<high) {
       int pi=partition(arr,low,high);
       quickSort(arr,low,pi-1);
       quickSort(arr,pi+1,high);
    }
}

public static void main(String[] args) {

    Order[] orders = {
       new Order(1,"Ram",3000),
       new Order(2,"John",1000),
```

```
        new Order(3,"Sam",5000)
    };

    quickSort(orders,0,orders.length-1);

    for(Order o:orders)
        System.out.println(o);
  }
}
```

Complexity

- Bubble Sort → $O(n^2)$
- Quick Sort → O(n log n)

Exercise 4: Employee Management System

Employee.java

```
public class Employee {
  int employeeId;
  String name;
  String position;
  double salary;

  public Employee(int id,String name,
            String position,double salary) {
    this.employeeId=id;
    this.name=name;
    this.position=position;
    this.salary=salary;
  }

  public String toString() {
    return employeeId+" "+name+
        " "+position+" "+salary;
  }
}
```

EmployeeManagement.java

```
public class EmployeeManagement {

  Employee[] employees = new Employee[10];
```

```
int count = 0;

void add(Employee e) {
   employees[count++] = e;
}

void search(int id) {
   for(int i=0;i<count;i++) {
      if(employees[i].employeeId==id)
         System.out.println(employees[i]);
   }
}

void traverse() {
   for(int i=0;i<count;i++)
      System.out.println(employees[i]);
}

void delete(int id) {
   for(int i=0;i<count;i++) {
      if(employees[i].employeeId==id) {
         for(int j=i;j<count-1;j++)
            employees[j]=employees[j+1];
         count--;
      }
   }
}

public static void main(String[] args) {
   EmployeeManagement obj =
         new EmployeeManagement();

   obj.add(new Employee(1,"Arun","Manager",50000));
   obj.add(new Employee(2,"Kumar","Developer",40000));

   obj.traverse();

   obj.delete(1);

   System.out.println("\nAfter Delete");
   obj.traverse();
```

```
    }
}
```

Exercise 5: Task Management System

Task.java

```
public class Task {
    int taskId;
    String taskName;
    String status;

    Task next;

    public Task(int id,String name,String status) {
        this.taskId=id;
        this.taskName=name;
        this.status=status;
    }
}
```

TaskManagement.java

```
public class TaskManagement {

    Task head;

    void add(int id,String name,String status) {
        Task t = new Task(id,name,status);

        if(head==null)
            head=t;
        else {
            Task temp=head;
            while(temp.next!=null)
                temp=temp.next;
            temp.next=t;
        }
    }

    void display() {
        Task temp=head;
```

```
        while(temp!=null) {
            System.out.println(
                temp.taskId+" "+
                temp.taskName+" "+
                temp.status);
            temp=temp.next;
        }
    }

    void delete(int id) {
        if(head.taskId==id) {
            head=head.next;
            return;
        }

        Task temp=head;
        while(temp.next!=null &&
              temp.next.taskId!=id)
            temp=temp.next;

        if(temp.next!=null)
            temp.next=temp.next.next;
    }

    public static void main(String[] args) {

        TaskManagement tm =
                new TaskManagement();

        tm.add(1,"Coding","Pending");
        tm.add(2,"Testing","Completed");

        tm.display();

        tm.delete(1);

        System.out.println("\nAfter Delete");
        tm.display();
    }
}
```

Exercise 6: Library Management System

Book.java

```
public class Book {
   int bookId;
   String title;
   String author;

   public Book(int id,String title,String author) {
      this.bookId=id;
      this.title=title;
      this.author=author;
   }
}
```

LibrarySearch.java

```
public class LibrarySearch {

   static int linearSearch(Book[] books,
                           String title) {

      for(int i=0;i<books.length;i++) {
         if(books[i].title.equalsIgnoreCase(title))
            return i;
      }
      return -1;
   }

   static int binarySearch(Book[] books,
                           String title) {

      int low=0,high=books.length-1;

      while(low<=high) {
         int mid=(low+high)/2;

         int result=
         books[mid].title.compareToIgnoreCase(title);

         if(result==0)
            return mid;
```

```
            else if(result<0)
                low=mid+1;
            else
                high=mid-1;
        }
        return -1;
    }

    public static void main(String[] args) {

        Book[] books = {
            new Book(1,"C","Author1"),
            new Book(2,"Java","Author2"),
            new Book(3,"Python","Author3")
        };

        System.out.println(
            linearSearch(books,"Java"));

        System.out.println(
            binarySearch(books,"Java"));
    }
}
```

Exercise 7: Financial Forecasting

FinancialForecast.java

```
public class FinancialForecast {

    static double futureValue(double amount,
                              double rate,
                              int years) {

        if(years==0)
            return amount;

        return futureValue(
                amount*(1+rate),
                rate,
                years-1);
    }
```

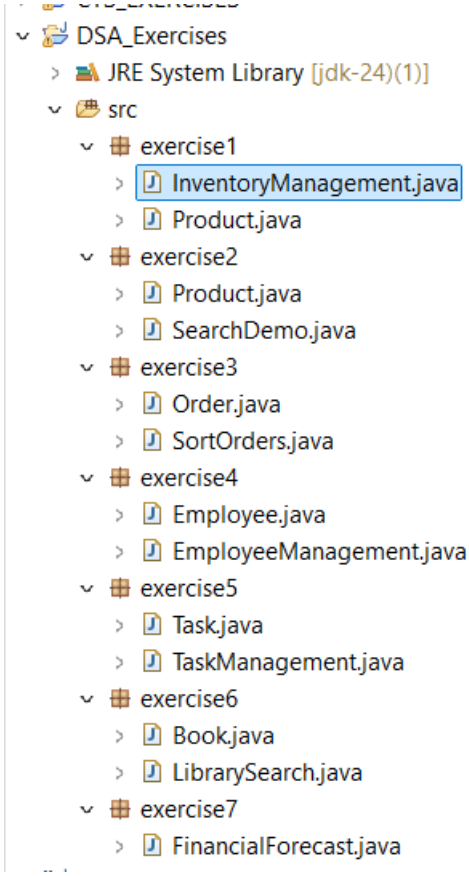
```
    public static void main(String[] args) {

        double result =
            futureValue(10000,0.10,5);

        System.out.println(
            "Future Value = " + result);
    }
}
```

Complexity

- Recursive approach → O(n)
- Space Complexity → O(n)


DSA_Exercises
JRE System Library [jdk-24)(1)]
src
exercise1
InventoryManagement.java
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exercise2
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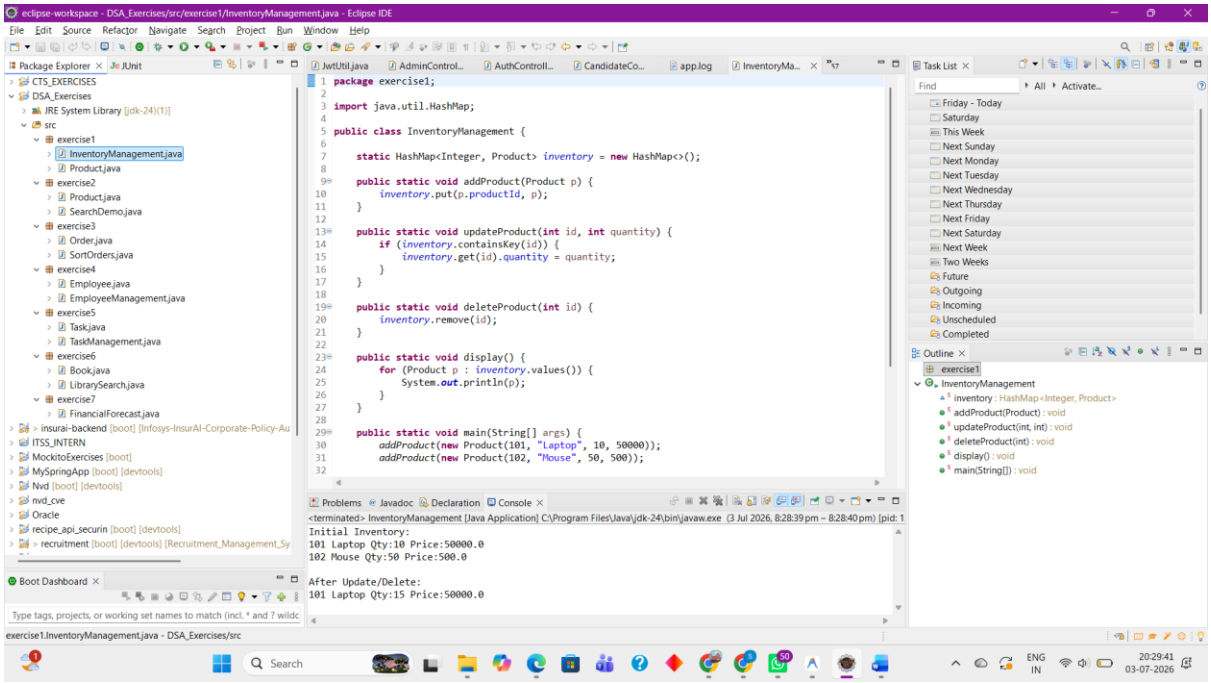
exercise3
Order.java
SortOrders.java
exercise4
Employee.java
EmployeeManagement.java
exercise5
Task.java
TaskManagement.java
exercise6
Book.java
LibrarySearch.java
exercise7
FinancialForecast.java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JRE System Library [jdk-24)(1)]
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exercise1
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exercise7
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JwtUtil.java
AdminControl...
AuthControll...
CandidateCo...
app.log
InventoryMa...
1 package exercise1;
2
3 import java.util.HashMap;
4
5 public class InventoryManagement {
6
7 static HashMap<Integer, Product> inventory = new HashMap<>();
8
9 public static void addProduct(Product p) {
10 inventory.put(p.productId, p);
11 }
12
13 public static void updateProduct(int id, int quantity) {
14 if (inventory.containsKey(id)) {
15 inventory.get(id).quantity = quantity;
16 }
17 }
18
19 public static void deleteProduct(int id) {
20 inventory.remove(id);
21 }
22
23 public static void display() {
24 for (Product p : inventory.values()) {
25 System.out.println(p);
26 }
27 }
28
29 public static void main(String[] args) {
30 addProduct(new Product(101, "Laptop", 10, 50000));
31 addProduct(new Product(102, "Mouse", 50, 500));
32
Problems
Javadoc
Declaration
Console
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Initial Inventory:
101 Laptop Qty:10 Price:50000.0
102 Mouse Qty:50 Price:500.0
After Update/Delete:
101 Laptop Qty:15 Price:50000.0
Task List
Find
All
Activate...
Friday - Today
Saturday
This Week
Next Sunday
Next Monday
Next Tuesday
Next Wednesday
Next Thursday
Next Friday
Next Saturday
Next Week
Two Weeks
Future
Outgoing
Incoming
Unscheduled
Completed
Outline
exercise1
InventoryManagement
inventory : HashMap<Integer, Product>
addProduct(Product) : void
updateProduct(int, int) : void
deleteProduct(int) : void
display() : void
main(String[]) : void
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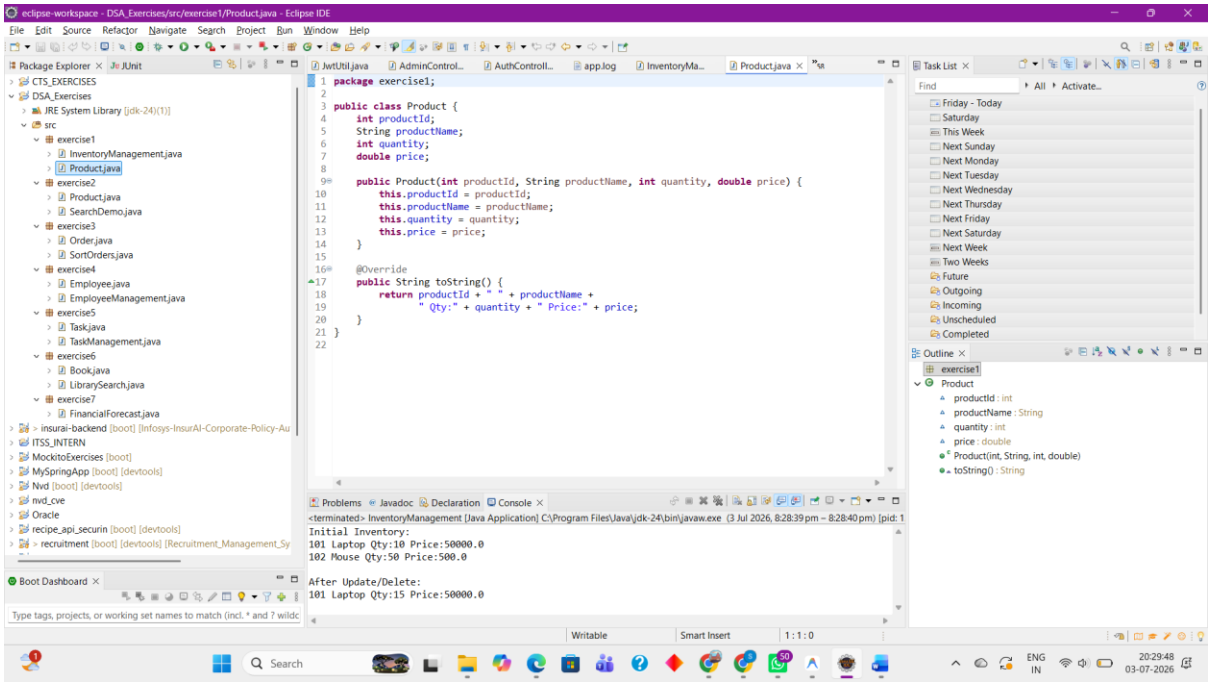
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JRE System Library [jdk-24)(1)]
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Product.java
1 package exercise1;
2
3 public class Product {
4 int productId;
5 String productName;
6 int quantity;
7 double price;
8
9 public Product(int productId, String productName, int quantity, double price) {
10 this.productId = productId;
11 this.productName = productName;
12 this.quantity = quantity;
13 this.price = price;
14 }
15
16 @Override
17 public String toString() {
18 return productId + " " + productName +
19 " Qty:" + quantity + " Price:" + price;
20 }
21 }
22
Problems
Javadoc
Declaration
Console
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All
Activate...
Friday - Today
Saturday
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Next Friday
Next Saturday
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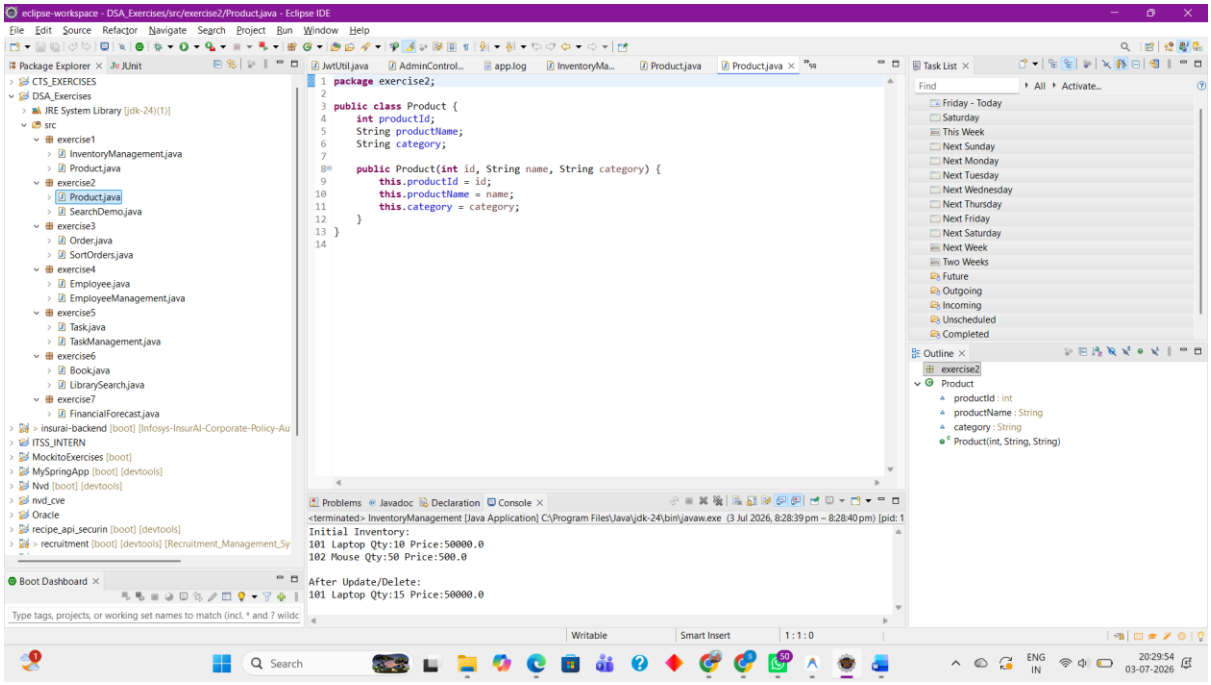
Two Weeks
Future
Outgoing
Incoming
Unscheduled
Completed
Outline
exercise1
Product
productId : int
productName : String
quantity : int
price : double
Product(int, String, int, double)
toString() : String
Writable
Smart Insert
1 : 1 : 0
Search
ENG
IN
20:29:48
03-07-2026


eclipse-workspace - DSA_Exercises/src/exercise2/Product.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
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CTS_EXERCISES
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JRE System Library [jdk-24)(1)]
src
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Product.java
Product.java
1 package exercise2;
2
3 public class Product {
4 int productId;
5 String productName;
6 String category;
7
8 public Product(int id, String name, String category) {
9 this.productId = id;
10 this.productName = name;
11 this.category = category;
12 }
13 }
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exercise2
Product
productId : int
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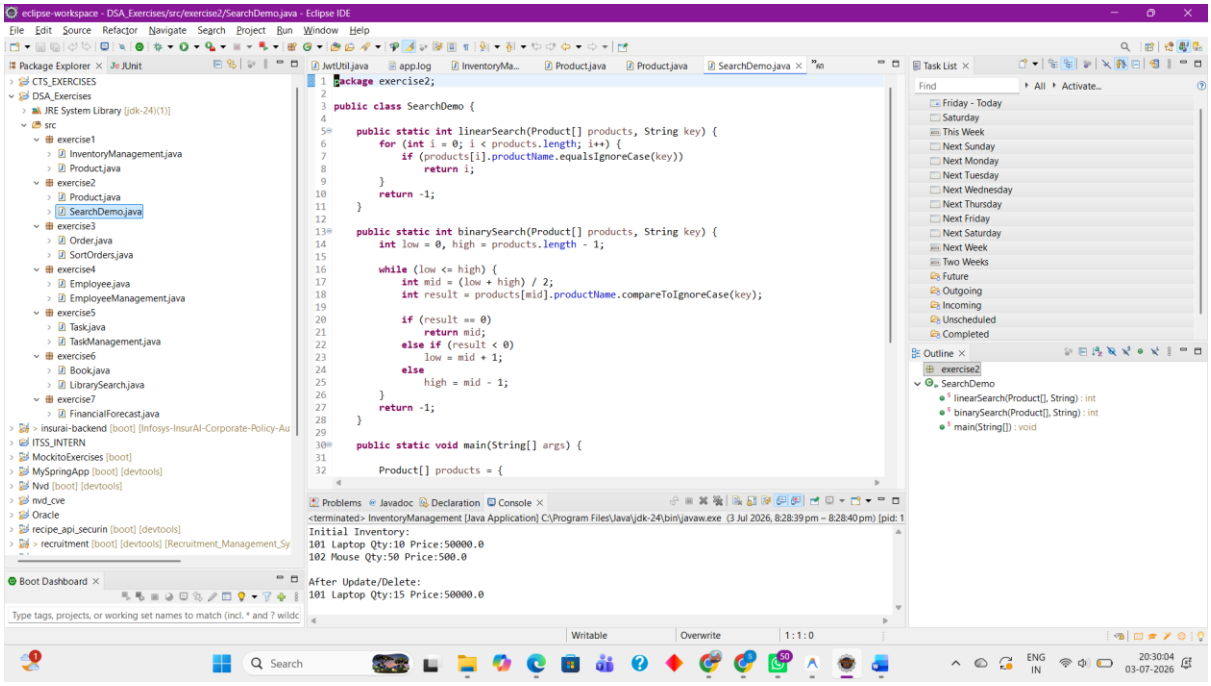
category : String
Product(int, String, String)
Writable
Smart Insert
1 : 1 : 0
Search
ENG
IN
20:29:54
03-07-2026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3 public class SearchDemo {
4
5 public static int linearSearch(Product[] products, String key) {
6 for (int i = 0; i < products.length; i++) {
7 if (products[i].productName.equalsIgnoreCase(key))
8 return i;
9 }
10 return -1;
11 }
12
13 public static int binarySearch(Product[] products, String key) {
14 int low = 0, high = products.length - 1;
15
16 while (low <= high) {
17 int mid = (low + high) / 2;
18 int result = products[mid].productName.compareToIgnoreCase(key);
19
20 if (result == 0)
21 return mid;
22 else if (result < 0)
23 low = mid + 1;
24 else
25 high = mid - 1;
26 }
27 return -1;
28 }
29
30 public static void main(String[] args) {
31
32 Product[] products = {
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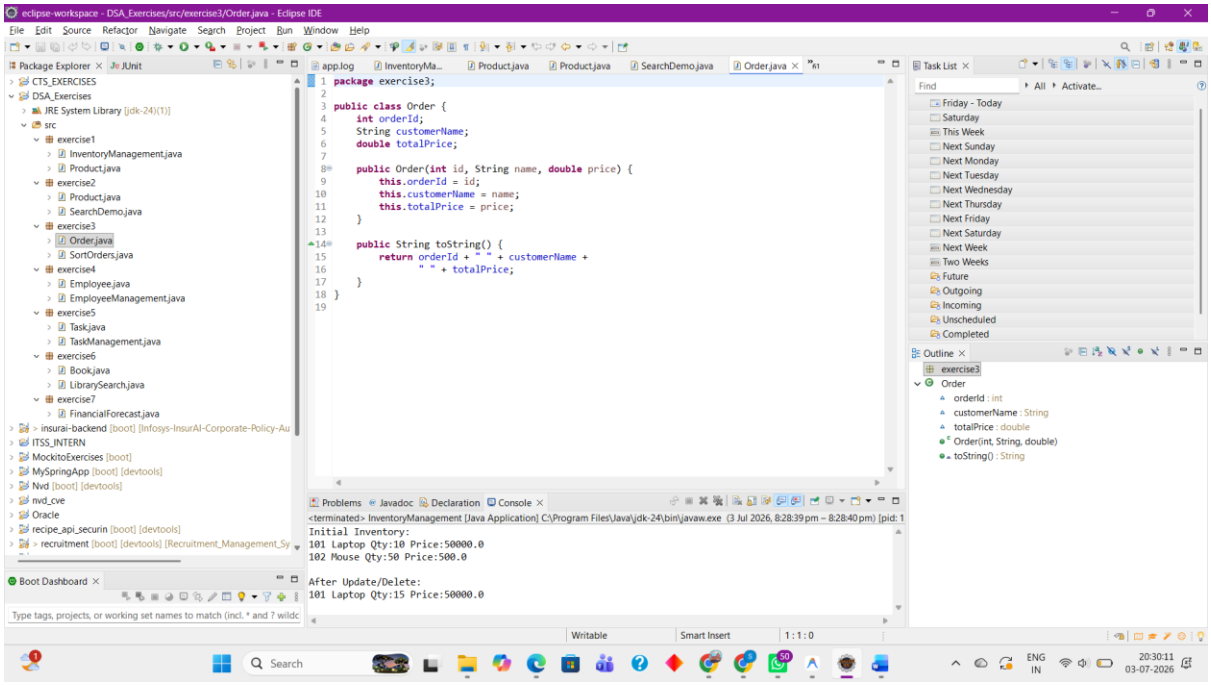
exercise2
SearchDemo
linearSearch(Product[], String) : int
binarySearch(Product[], String) : int
main(String[]) : void
Writable
Overwrite
1 : 1 : 0
Search
ENG
IN
20:30:04
03-07-2026


eclipse-workspace - DSA_Exercises/src/exercise3/Order.java - Eclipse IDE
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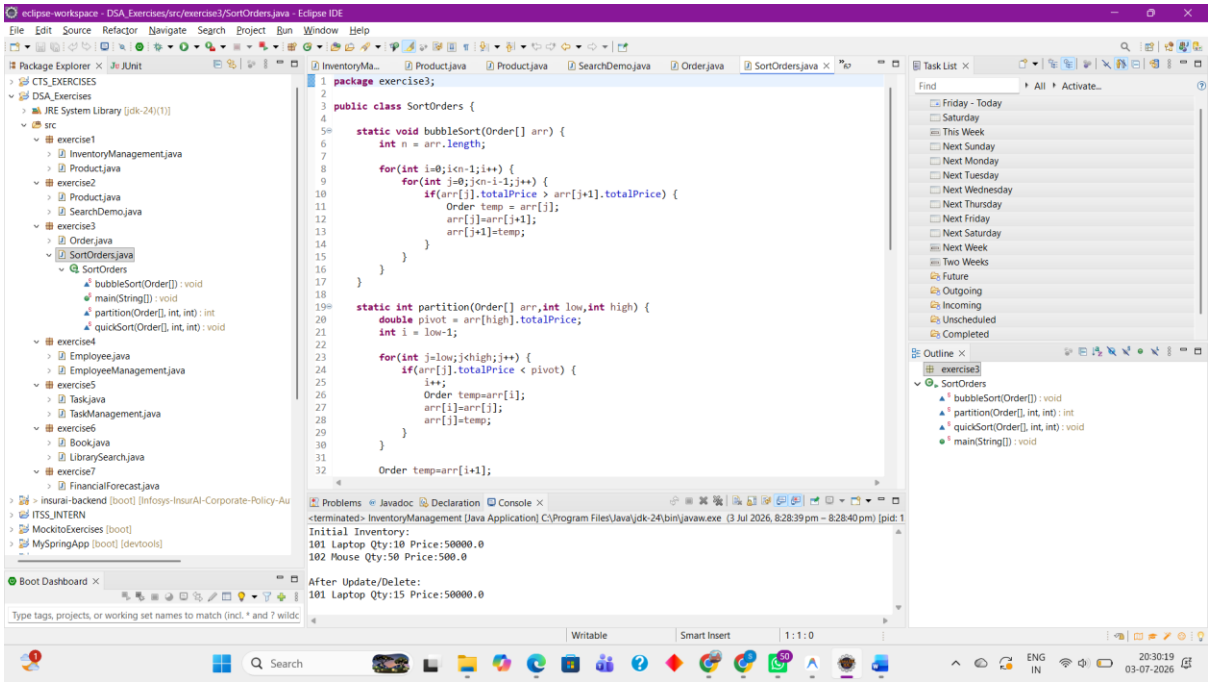
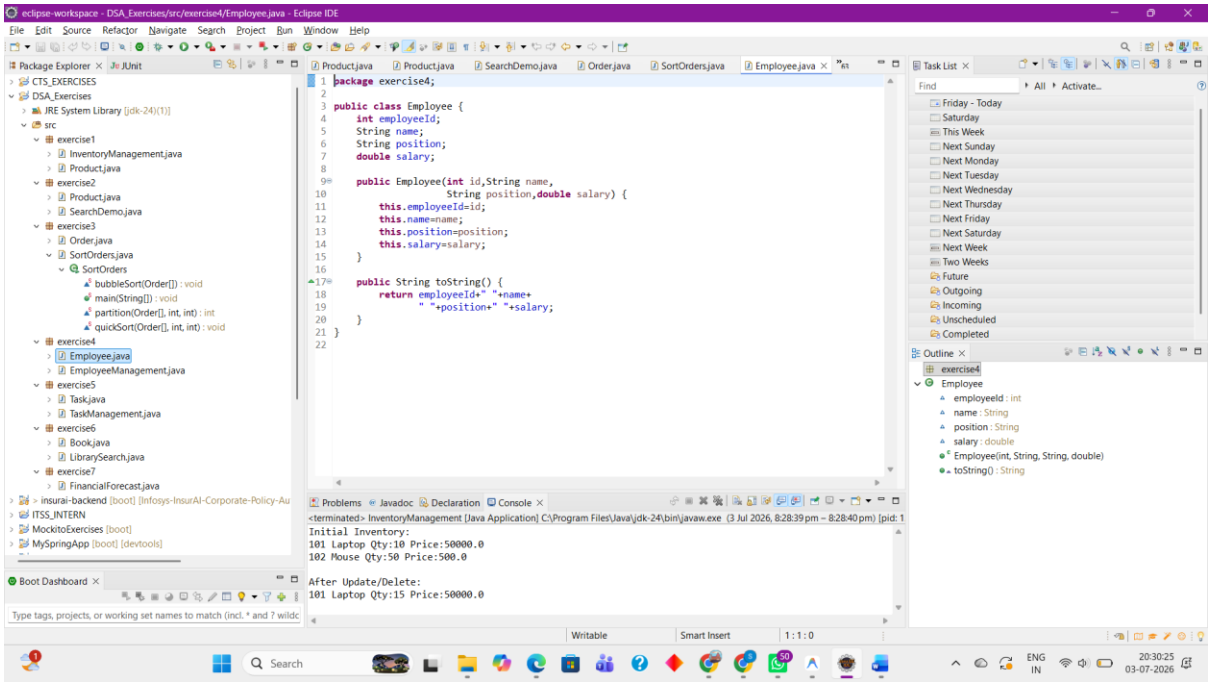
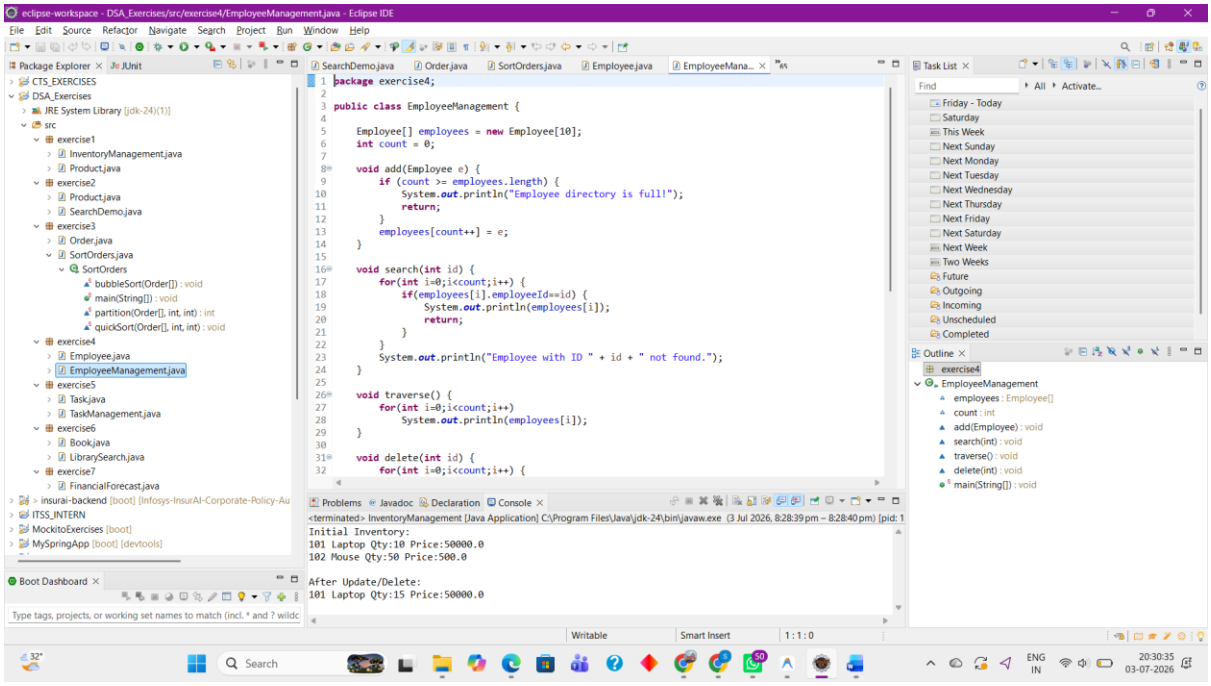
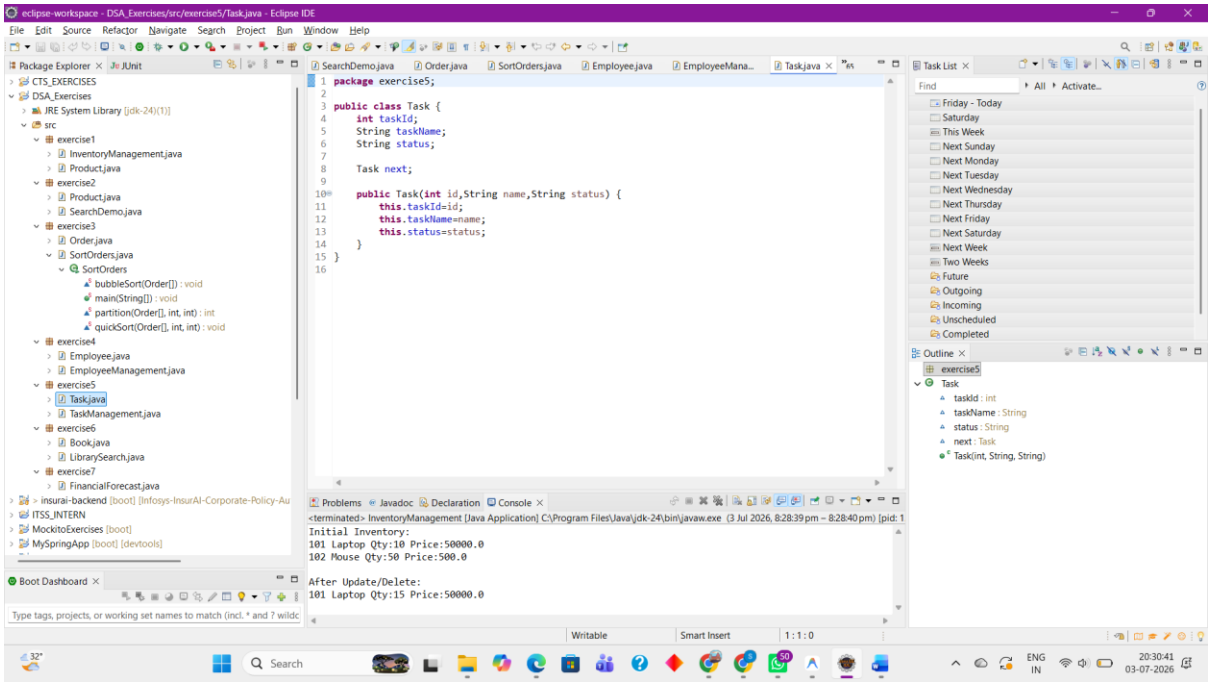
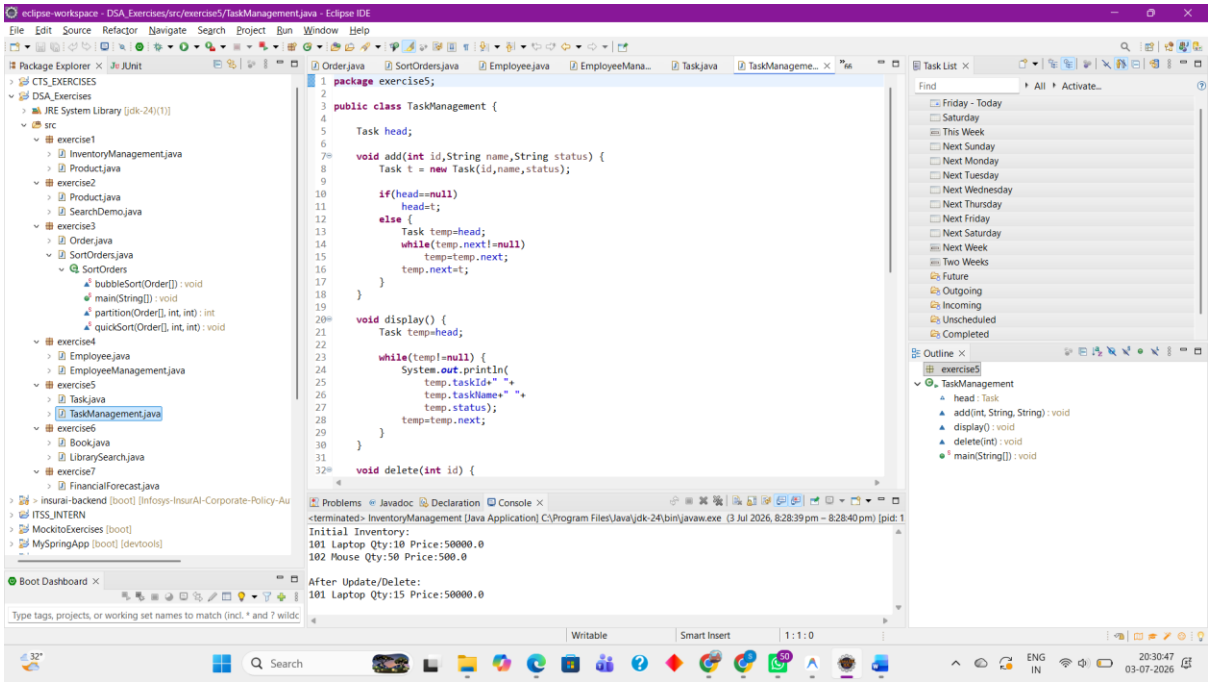
Product.java
SearchDemo.java
Order.java
1 package exercise3;
2
3 public class Order {
4 int orderId;
5 String customerName;
6 double totalPrice;
7
8 public Order(int id, String name, double price) {
9 this.orderId = id;
10 this.customerName = name;
11 this.totalPrice = price;
12 }
13
14 public String toString() {
15 return orderId + " " + customerName +
16 " " + totalPrice;
17 }
18 }
19
Problems
Javadoc
Declaration
Console
<terminated> InventoryManagement [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (3 Jul 2026, 8:28:39 pm – 8:28:40 pm) [pid: 1
Initial Inventory:
101 Laptop Qty:10 Price:50000.0
102 Mouse Qty:50 Price:500.0
After Update/Delete:
101 Laptop Qty:15 Price:50000.0
Task List
Find
All
Activate...
Friday - Today
Saturday
This Week
Next Sunday
Next Monday
Next Tuesday
Next Wednesday
Next Thursday
Next Friday
Next Saturday
Next Week
Two Weeks
Future
Outgoing
Incoming
Unscheduled
Completed
Outline
exercise3
Order
orderId : int
customerName : String
totalPrice : double
Order(int, String, double)
toString() : String
Writable
Smart Insert
1 : 1 : 0
Search
ENG
IN
20:30:11
03-07-2026

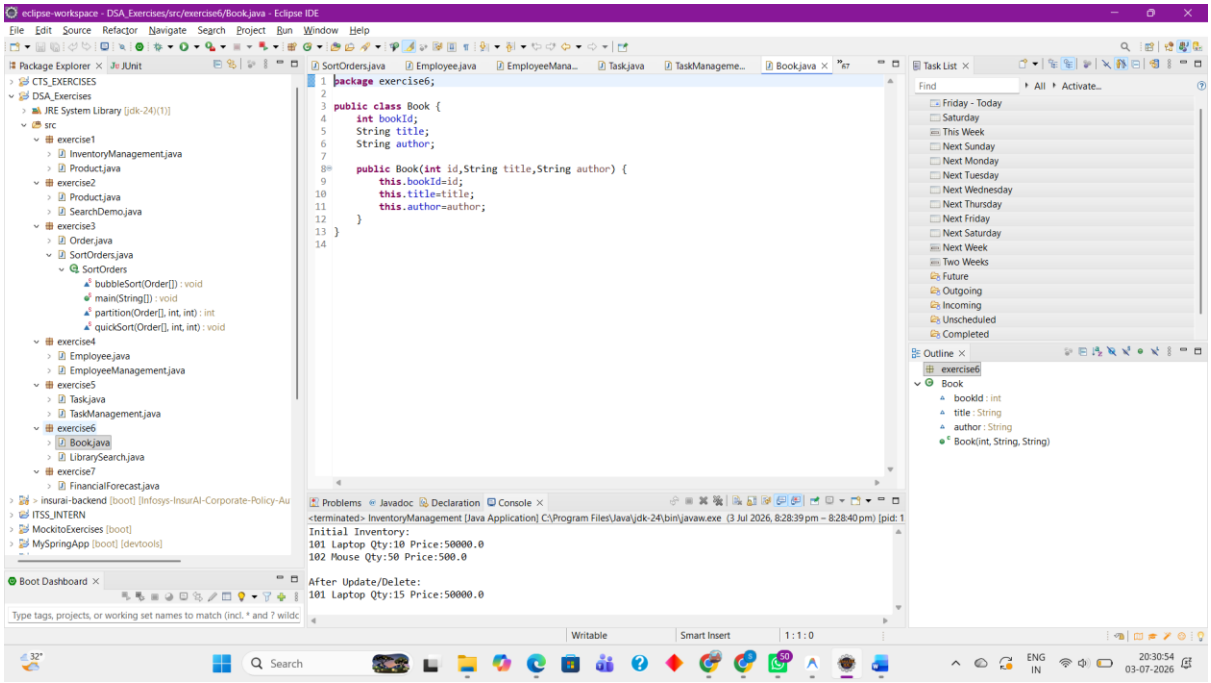

eclipse-workspace - DSA_Exercises/src/exercise3/SortOrders.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
JUnit
CTS_EXERCISES
DSA_Exercises
JRE System Library [jdk-24)(1)]
src
exercise1
InventoryManagement.java
Product.java
exercise2
Product.java
SearchDemo.java
exercise3
Order.java
SortOrders.java
SortOrders
bubbleSort(Order[]) : void
main(String[]) : void
partition(Order[], int, int) : int
quickSort(Order[], int, int) : void
exercise4
Employee.java
EmployeeManagement.java
exercise5
Task.java
TaskManagement.java
exercise6
Book.java
LibrarySearch.java
exercise7
FinancialForecast.java
insurai-backend [boot] [Infosys-InsurAI-Corporate-Policy-Au
ITSS_INTERN
MockitoExercises [boot]
MySpringApp [boot] [devtools]
Boot Dashboard
InventoryMa...
Product.java
Product.java
SearchDemo.java
Order.java
SortOrders.java
1 package exercise3;
2
3 public class SortOrders {
4
5 static void bubbleSort(Order[] arr) {
6 int n = arr.length;
7
8 for(int i=0;i<n-1;i++) {
9 for(int j=0;j<n-i-1;j++) {
10 if(arr[j].totalPrice > arr[j+1].totalPrice) {
11 Order temp = arr[j];
12 arr[j]=arr[j+1];
13 arr[j+1]=temp;
14 }
15 }
16 }
17 }
18
19 static int partition(Order[] arr,int low,int high) {
20 double pivot = arr[high].totalPrice;
21 int i = low-1;
22
23 for(int j=low;j<high;j++) {
24 if(arr[j].totalPrice < pivot) {
25 i++;
26 Order temp=arr[i];
27 arr[i]=arr[j];
28 arr[j]=temp;
29 }
30 }
31
32 Order temp=arr[i+1];
Problems
Javadoc
Declaration
Console
<terminated> InventoryManagement [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (3 Jul 2026, 8:28:39 pm – 8:28:40 pm) [pid: 1
Initial Inventory:
101 Laptop Qty:10 Price:50000.0
102 Mouse Qty:50 Price:500.0
After Update/Delete:
101 Laptop Qty:15 Price:50000.0
Task List
Find
All
Activate...
Friday - Today
Saturday
This Week
Next Sunday
Next Monday
Next Tuesday
Next Wednesday
Next Thursday
Next Friday
Next Saturday
Next Week
Two Weeks
Future
Outgoing
Incoming
Unscheduled
Completed
Outline
exercise3
SortOrders
bubbleSort(Order[]) : void
partition(Order[], int, int) : int
quickSort(Order[], int, int) : void
main(String[]) : void
Writable
Smart Insert
1 : 1 : 0
Search
ENG
IN
20:30:19
03-07-2026


eclipse-workspace - DSA_Exercises/src/exercise4/Employee.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
JUnit
CTS_EXERCISES
DSA_Exercises
JRE System Library [jdk-24)(1)]
src
exercise1
InventoryManagement.java
Product.java
exercise2
Product.java
SearchDemo.java
exercise3
Order.java
SortOrders.java
SortOrders
bubbleSort(Order[]) : void
main(String[]) : void
partition(Order[], int, int) : int
quickSort(Order[], int, int) : void
exercise4
Employee.java
EmployeeManagement.java
exercise5
Task.java
TaskManagement.java
exercise6
Book.java
LibrarySearch.java
exercise7
FinancialForecast.java
insurai-backend [boot] [Infosys-InsurAI-Corporate-Policy-Au
ITSS_INTERN
MockitoExercises [boot]
MySpringApp [boot] [devtools]
Boot Dashboard
Product.java
Product.java
SearchDemo.java
Order.java
SortOrders.java
Employee.java
1 package exercise4;
2
3 public class Employee {
4 int employeeId;
5 String name;
6 String position;
7 double salary;
8
9 public Employee(int id,String name,
10 String position,double salary) {
11 this.employeeId=id;
12 this.name=name;
13 this.position=position;
14 this.salary=salary;
15 }
16
17 public String toString() {
18 return employeeId+" "+name+
19 " "+position+" "+salary;
20 }
21 }
22
Problems
Javadoc
Declaration
Console
<terminated> InventoryManagement [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (3 Jul 2026, 8:28:39 pm – 8:28:40 pm) [pid: 1
Initial Inventory:
101 Laptop Qty:10 Price:50000.0
102 Mouse Qty:50 Price:500.0
After Update/Delete:
101 Laptop Qty:15 Price:50000.0
Task List
Find
All
Activate...
Friday - Today
Saturday
This Week
Next Sunday
Next Monday
Next Tuesday
Next Wednesday
Next Thursday
Next Friday
Next Saturday
Next Week
Two Weeks
Future
Outgoing
Incoming
Unscheduled
Completed
Outline
exercise4
Employee
employeeId : int
name : String
position : String
salary : double
Employee(int, String, String, double)
toString() : String
Writable
Smart Insert
1 : 1 : 0
Search
ENG
IN
20:30:25
03-07-2026


eclipse-workspace - DSA_Exercises/src/exercise4/EmployeeManagement.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JUnit
CTS_EXERCISES
DSA_Exercises
JRE System Library [jdk-24)(1)]
src
exercise1
InventoryManagement.java
Product.java
exercise2
Product.java
SearchDemo.java
exercise3
Order.java
SortOrders.java
SortOrders
bubbleSort(Order[]) : void
main(String[]) : void
partition(Order[], int, int) : int
quickSort(Order[], int, int) : void
exercise4
Employee.java
EmployeeManagement.java
exercise5
Task.java
TaskManagement.java
exercise6
Book.java
LibrarySearch.java
exercise7
FinancialForecast.java
insurai-backend [boot] [Infosys-InsurAI-Corporate-Policy-Au
ITSS_INTERN
MockitoExercises [boot]
MySpringApp [boot] [devtools]
Boot Dashboard
SearchDemo.java
Order.java
SortOrders.java
Employee.java
EmployeeMana...
1 package exercise4;
2
3 public class EmployeeManagement {
4
5 Employee[] employees = new Employee[10];
6 int count = 0;
7
8 void add(Employee e) {
9 if (count >= employees.length) {
10 System.out.println("Employee directory is full!");
11 return;
12 }
13 employees[count++] = e;
14 }
15
16 void search(int id) {
17 for(int i=0;i<count;i++) {
18 if(employees[i].employeeId==id) {
19 System.out.println(employees[i]);
20 return;
21 }
22 }
23 System.out.println("Employee with ID " + id + " not found.");
24 }
25
26 void traverse() {
27 for(int i=0;i<count;i++)
28 System.out.println(employees[i]);
29 }
30
31 void delete(int id) {
32 for(int i=0;i<count;i++) {
Problems Javadoc Declaration Console
<terminated> InventoryManagement [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (3 Jul 2026, 8:28:39 pm – 8:28:40 pm) [pid: 1
Initial Inventory:
101 Laptop Qty:10 Price:50000.0
102 Mouse Qty:50 Price:500.0
After Update/Delete:
101 Laptop Qty:15 Price:50000.0
Task List
Find All Activate...
Friday - Today
Saturday
This Week
Next Sunday
Next Monday
Next Tuesday
Next Wednesday
Next Thursday
Next Friday
Next Saturday
Next Week
Two Weeks
Future
Outgoing
Incoming
Unscheduled
Completed
Outline
exercise4
EmployeeManagement
employees : Employee[]
count : int
add(Employee) : void
search(int) : void
traverse() : void
delete(int) : void
main(String[]) : void
Writable
Smart Insert
1 : 1 : 0
Search
ENG
IN
20:30:35
03-07-2026


eclipse-workspace - DSA_Exercises/src/exercise5/Task.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JUnit
CTS_EXERCISES
DSA_Exercises
JRE System Library [jdk-24)(1)]
src
exercise1
InventoryManagement.java
Product.java
exercise2
Product.java
SearchDemo.java
exercise3
Order.java
SortOrders.java
SortOrders
bubbleSort(Order[]) : void
main(String[]) : void
partition(Order[], int, int) : int
quickSort(Order[], int, int) : void
exercise4
Employee.java
EmployeeManagement.java
exercise5
Task.java
TaskManagement.java
exercise6
Book.java
LibrarySearch.java
exercise7
FinancialForecast.java
insurai-backend [boot] [Infosys-InsurAI-Corporate-Policy-Au
ITSS_INTERN
MockitoExercises [boot]
MySpringApp [boot] [devtools]
Boot Dashboard
SearchDemo.java
Order.java
SortOrders.java
Employee.java
EmployeeMana...
Task.java
1 package exercise5;
2
3 public class Task {
4 int taskId;
5 String taskName;
6 String status;
7
8 Task next;
9
10 public Task(int id,String name,String status) {
11 this.taskId=id;
12 this.taskName=name;
13 this.status=status;
14 }
15 }
16
Problems Javadoc Declaration Console
<terminated> InventoryManagement [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (3 Jul 2026, 8:28:39 pm – 8:28:40 pm) [pid: 1
Initial Inventory:
101 Laptop Qty:10 Price:50000.0
102 Mouse Qty:50 Price:500.0
After Update/Delete:
101 Laptop Qty:15 Price:50000.0
Task List
Find All Activate...
Friday - Today
Saturday
This Week
Next Sunday
Next Monday
Next Tuesday
Next Wednesday
Next Thursday
Next Friday
Next Saturday
Next Week
Two Weeks
Future
Outgoing
Incoming
Unscheduled
Completed
Outline
exercise5
Task
taskId : int
taskName : String
status : String
next : Task
Task(int, String, String)
Writable
Smart Insert
1 : 1 : 0
Search
ENG
IN
20:30:41
03-07-2026


eclipse-workspace - DSA_Exercises/src/exercise5/TaskManagement.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
JUnit
CTS_EXERCISES
DSA_Exercises
JRE System Library [jdk-24)(1)]
src
exercise1
InventoryManagement.java
Product.java
exercise2
Product.java
SearchDemo.java
exercise3
Order.java
SortOrders.java
SortOrders
bubbleSort(Order[]) : void
main(String[]) : void
partition(Order[], int, int) : int
quickSort(Order[], int, int) : void
exercise4
Employee.java
EmployeeManagement.java
exercise5
Task.java
TaskManagement.java
exercise6
Book.java
LibrarySearch.java
exercise7
FinancialForecast.java
insurai-backend [boot] [Infosys-InsurAI-Corporate-Policy-Au
ITSS_INTERN
MockitoExercises [boot]
MySpringApp [boot] [devtools]
Boot Dashboard
Order.java
SortOrders.java
Employee.java
EmployeeMana...
Task.java
TaskManageme...
1 package exercise5;
2
3 public class TaskManagement {
4
5 Task head;
6
7 void add(int id,String name,String status) {
8 Task t = new Task(id,name,status);
9
10 if(head==null)
11 head=t;
12 else {
13 Task temp=head;
14 while(temp.next!=null)
15 temp=temp.next;
16 temp.next=t;
17 }
18 }
19
20 void display() {
21 Task temp=head;
22
23 while(temp!=null) {
24 System.out.println(
25 temp.taskId+" "+
26 temp.taskName+" "+
27 temp.status);
28 temp=temp.next;
29 }
30 }
31
32 void delete(int id) {
Problems Javadoc Declaration Console
<terminated> InventoryManagement [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (3 Jul 2026, 8:28:39 pm – 8:28:40 pm) [pid: 1
Initial Inventory:
101 Laptop Qty:10 Price:50000.0
102 Mouse Qty:50 Price:500.0
After Update/Delete:
101 Laptop Qty:15 Price:50000.0
Task List
Find
All
Activate...
Friday - Today
Saturday
This Week
Next Sunday
Next Monday
Next Tuesday
Next Wednesday
Next Thursday
Next Friday
Next Saturday
Next Week
Two Weeks
Future
Outgoing
Incoming
Unscheduled
Completed
Outline
exercise5
TaskManagement
head : Task
add(int, String, String) : void
display() : void
delete(int) : void
main(String[]) : void
Writable
Smart Insert
1 : 1 : 0
Search
ENG
IN
20:30:47
03-07-2026


eclipse-workspace - DSA_Exercises/src/exercise6/Book.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
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CTS_EXERCISES
DSA_Exercises
JRE System Library [jdk-24)(1)]
src
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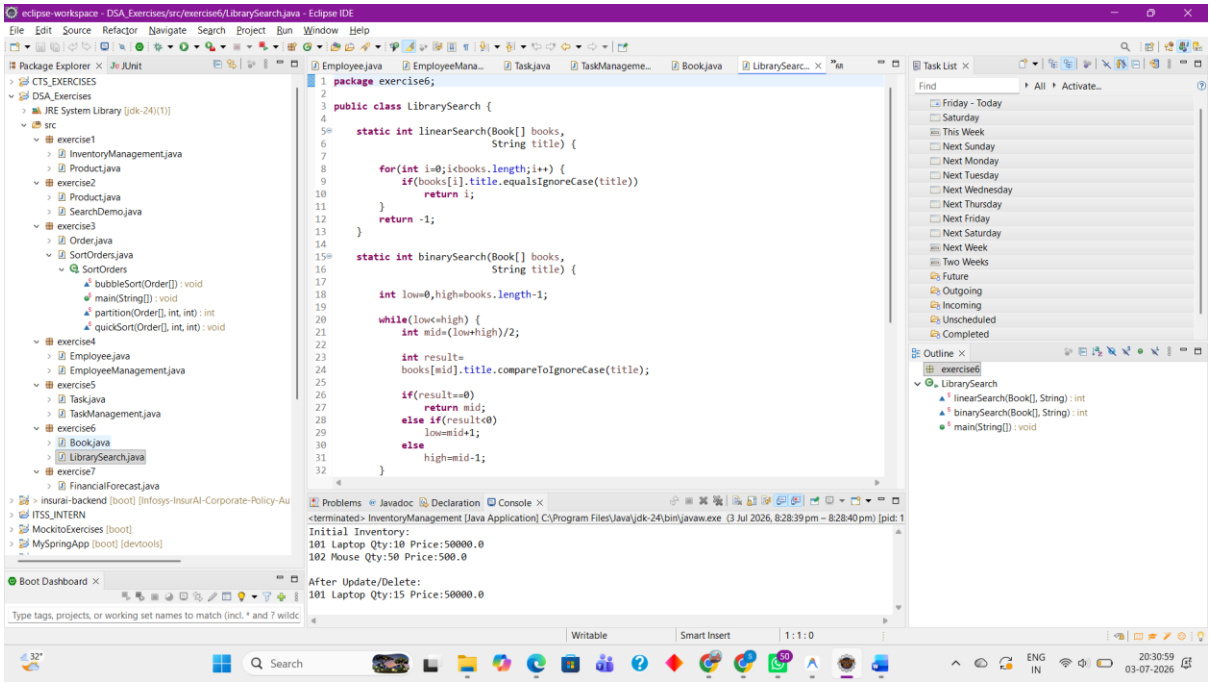
InventoryManagement.java
Product.java
exercise2
Product.java
SearchDemo.java
exercise3
Order.java
SortOrders.java
SortOrders
bubbleSort(Order[]) : void
main(String[]) : void
partition(Order[], int, int) : int
quickSort(Order[], int, int) : void
exercise4
Employee.java
EmployeeManagement.java
exercise5
Task.java
TaskManagement.java
exercise6
Book.java
LibrarySearch.java
exercise7
FinancialForecast.java
insurai-backend [boot] [Infosys-InsurAI-Corporate-Policy-Au
ITSS_INTERN
MockitoExercises [boot]
MySpringApp [boot] [devtools]
Boot Dashboard
SortOrders.java
Employee.java
EmployeeMana...
Task.java
TaskManageme...
Book.java
1 package exercise6;
2
3 public class Book {
4 int bookId;
5 String title;
6 String author;
7
8 public Book(int id,String title,String author) {
9 this.bookId=id;
10 this.title=title;
11 this.author=author;
12 }
13 }
14
Problems Javadoc Declaration Console
<terminated> InventoryManagement [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (3 Jul 2026, 8:28:39 pm – 8:28:40 pm) [pid: 1
Initial Inventory:
101 Laptop Qty:10 Price:50000.0
102 Mouse Qty:50 Price:500.0
After Update/Delete:
101 Laptop Qty:15 Price:50000.0
Task List
Find
All
Activate...
Friday - Today
Saturday
This Week
Next Sunday
Next Monday
Next Tuesday
Next Wednesday
Next Thursday
Next Friday
Next Saturday
Next Week
Two Weeks
Future
Outgoing
Incoming
Unscheduled
Completed
Outline
exercise6
Book
bookId : int
title : String
author : String
Book(int, String, String)
Writable
Smart Insert
1 : 1 : 0
Search
ENG
IN
20:30:54
03-07-2026


eclipse-workspace - DSA_Exercises/src/exercise6/LibrarySearch.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
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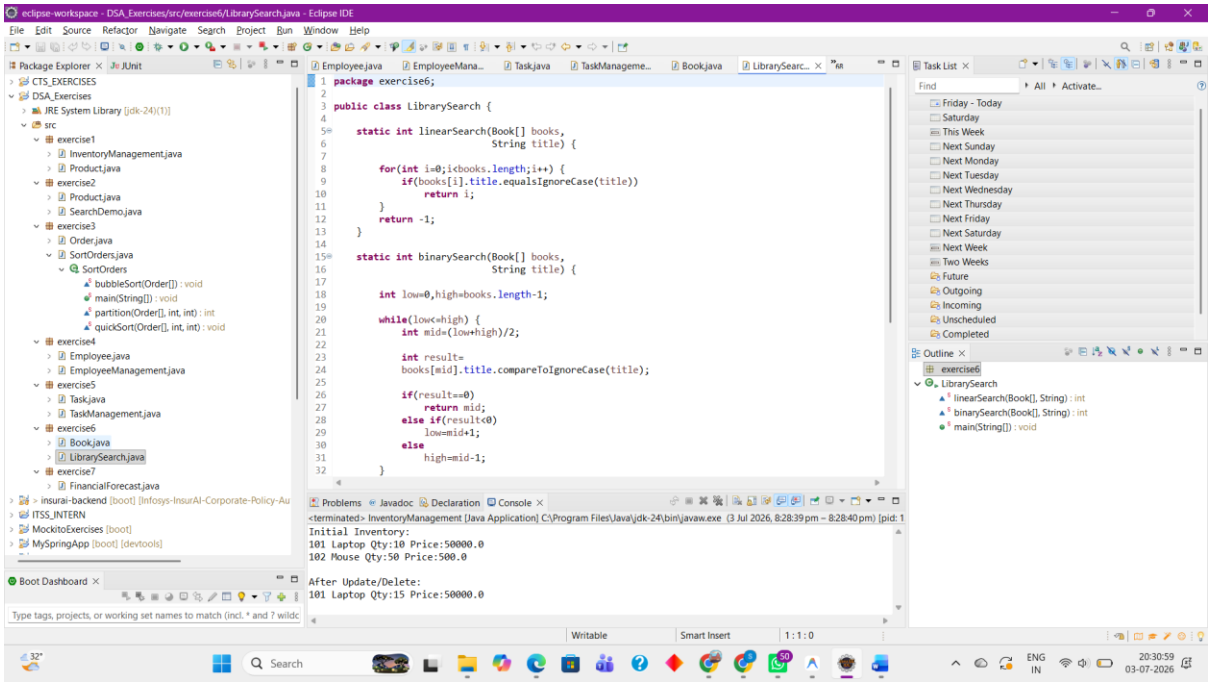
CTS_EXERCISES
DSA_Exercises
JRE System Library [jdk-24)(1)]
src
exercise1
InventoryManagement.java
Product.java
exercise2
Product.java
SearchDemo.java
exercise3
Order.java
SortOrders.java
SortOrders
bubbleSort(Order[]) : void
main(String[]) : void
partition(Order[], int, int) : int
quickSort(Order[], int, int) : void
exercise4
Employee.java
EmployeeManagement.java
exercise5
Task.java
TaskManagement.java
exercise6
Book.java
LibrarySearch.java
exercise7
FinancialForecast.java
insurai-backend [boot] [Infosys-InsurAI-Corporate-Policy-Au
ITSS_INTERN
MockitoExercises [boot]
MySpringApp [boot] [devtools]
Boot Dashboard
Type tags, projects, or working set names to match (incl. * and ? wildc
Employee.java
EmployeeMana...
Task.java
TaskManageme...
Book.java
LibrarySearc...
1 package exercise6;
2
3 public class LibrarySearch {
4
5 static int linearSearch(Book[] books,
6 String title) {
7
8 for(int i=0;i<books.length;i++) {
9 if(books[i].title.equalsIgnoreCase(title))
10 return i;
11 }
12 return -1;
13 }
14
15 static int binarySearch(Book[] books,
16 String title) {
17
18 int low=0,high=books.length-1;
19
20 while(low<=high) {
21 int mid=(low+high)/2;
22
23 int result=
24 books[mid].title.compareToIgnoreCase(title);
25
26 if(result==0)
27 return mid;
28 else if(result<0)
29 low=mid+1;
30 else
31 high=mid-1;
32 }
Problems
Javadoc
Declaration
Console
<terminated> InventoryManagement [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (3 Jul 2026, 8:28:39 pm – 8:28:40 pm) [pid: 1
Initial Inventory:
101 Laptop Qty:10 Price:50000.0
102 Mouse Qty:50 Price:500.0
After Update/Delete:
101 Laptop Qty:15 Price:50000.0
Task List
Find
All
Activate...
Friday - Today
Saturday
This Week
Next Sunday
Next Monday
Next Tuesday
Next Wednesday
Next Thursday
Next Friday
Next Saturday
Next Week
Two Weeks
Future
Outgoing
Incoming
Unscheduled
Completed
Outline
exercise6
LibrarySearch
linearSearch(Book[], String) : int
binarySearch(Book[], String) : int
main(String[]) : void
Writable
Smart Insert
1 : 1 : 0
Search
ENG
IN
20:30:59
03-07-2026


eclipse-workspace - DSA_Exercises/src/exercise6/LibrarySearch.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
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CTS_EXERCISES
DSA_Exercises
JRE System Library [jdk-24)(1)]
src
exercise1
InventoryManagement.java
Product.java
exercise2
Product.java
SearchDemo.java
exercise3
Order.java
SortOrders.java
SortOrders
bubbleSort(Order[]) : void
main(String[]) : void
partition(Order[], int, int) : int
quickSort(Order[], int, int) : void
exercise4
Employee.java
EmployeeManagement.java
exercise5
Task.java
TaskManagement.java
exercise6
Book.java
LibrarySearch.java
exercise7
FinancialForecast.java
insurai-backend [boot] [Infosys-InsurAI-Corporate-Policy-Au
ITSS_INTERN
MockitoExercises [boot]
MySpringApp [boot] [devtools]
Boot Dashboard
Type tags, projects, or working set names to match (incl. * and ? wildc
Employee.java
EmployeeMana...
Task.java
TaskManageme...
Book.java
LibrarySearc...
1 package exercise6;
2
3 public class LibrarySearch {
4
5 static int linearSearch(Book[] books,
6 String title) {
7
8 for(int i=0;i<books.length;i++) {
9 if(books[i].title.equalsIgnoreCase(title))
10 return i;
11 }
12 return -1;
13 }
14
15 static int binarySearch(Book[] books,
16 String title) {
17
18 int low=0,high=books.length-1;
19
20 while(low<=high) {
21 int mid=(low+high)/2;
22
23 int result=
24 books[mid].title.compareToIgnoreCase(title);
25
26 if(result==0)
27 return mid;
28 else if(result<0)
29 low=mid+1;
30 else
31 high=mid-1;
32 }
Problems
Javadoc
Declaration
Console
<terminated> InventoryManagement [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (3 Jul 2026, 8:28:39 pm – 8:28:40 pm) [pid: 1
Initial Inventory:
101 Laptop Qty:10 Price:50000.0
102 Mouse Qty:50 Price:500.0
After Update/Delete:
101 Laptop Qty:15 Price:50000.0
Task List
Find
All
Activate...
Friday - Today
Saturday
This Week
Next Sunday
Next Monday
Next Tuesday
Next Wednesday
Next Thursday
Next Friday
Next Saturday
Next Week
Two Weeks
Future
Outgoing
Incoming
Unscheduled
Completed
Outline
exercise6
LibrarySearch
linearSearch(Book[], String) : int
binarySearch(Book[], String) : int
main(String[]) : void
Writable
Smart Insert
1 : 1 : 0
Search
ENG
IN
20:30:59
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